WEEK 4 ASSESSMENT

Data + ML Fundamentals Review

Quiz • Preprocessing Task • Concept Viva

Date	28-Oct-2026	**Duration**	4 hours
Difficulty	Intermediate	**Total marks**	100

Assessment Instructions

- Complete independently. Use notes only if the trainer permits them.
- Write short explanations where requested and do not copy code without understanding it.
- You may be asked to explain any line of your notebook during the viva.
- Submit quiz answers and your preprocessing notebook.
- Suggested timing: Quiz 45 minutes; practical task 2 hours; debugging, reflection and viva 1 hour 15 minutes.

Part A — Multiple-Choice Quiz (20 marks)

Choose one best answer for each question. Each question carries 1 mark.

Which statement best describes Machine Learning?

- **A. A system learns patterns from examples**
- B. A program only follows fixed rules
- C. A database stores files
- D. A network connects devices

Predicting whether a transaction is fraudulent or legitimate is usually:

- A. Regression
- **B. Classification**
- C. Clustering
- D. Data compression

Predicting a numeric house selling price is usually:

- A. Classification
- B. Clustering
- **C. Regression**
- D. One-hot encoding

Grouping customers by behaviour without existing group labels is:

- A. Supervised regression

- B. Supervised classification
- **C. Unsupervised clustering**
- D. Scaling

In a dataset used to predict churn, the column churn is the:

- A. Scaler
- **B. Label / target**
- C. Test set
- D. Imputer

Which is a possible feature for predicting house price?

- A. The price to be predicted
- **B. Number of bedrooms**
- C. The model's final score
- D. The prediction itself

Why keep a test set separate?

- A. To enlarge the dataset
- **B. To estimate performance on unseen data**
- C. To fill missing values
- D. To encode categories

A missing-value imputer is used for:

- **A. Handling missing entries**
- B. Splitting rows
- C. Choosing the target automatically
- D. Measuring accuracy

A common strategy for a missing numeric feature is:

- A. Replace values with the column name
- **B. Use the median learned from training data**
- C. Copy values from the target
- D. Convert numbers to city names

StandardScaler generally:

- A. Converts categories to text
- **B. Standardises numeric values using learned statistics**
- C. Deletes test rows

- D. Predicts labels

A common method to encode nominal data such as city is:

- **A. One-hot encoding**
- B. Median imputation
- C. Train/test split
- D. Accuracy scoring

OneHotEncoder(handle_unknown='ignore') helps when:

- A. All values are missing
- **B. A category appears during transform that was not seen during fitting**
- C. There are no features
- D. A model overfits

Data leakage occurs when:

- A. A notebook is not saved
- **B. Held-out or unavailable information improperly influences training/evaluation**
- C. A dataset has many columns
- D. Accuracy is low

Imputers and scalers should generally be fitted:

- A. On all rows before splitting
- **B. On the test set only**
- C. On the training set only
- D. On unrelated data

In scikit-learn, fit usually means:

- **A. Learn parameters or rules from data**
- B. Display a chart
- C. Remove the target automatically
- D. Save a notebook

In scikit-learn, transform usually means:

- **A. Apply a learned transformation**
- B. Learn from test labels
- C. Create business requirements
- D. Split rows randomly

A benefit of a scikit-learn Pipeline is that it:

- A. Guarantees 100% accuracy
- **B. Combines steps and applies them consistently**
- C. Removes evaluation needs
- D. Makes all missing values harmless

Which problem may be better solved with a simple rule?

- **A. Flag an invoice number that duplicates one for the same vendor**
- B. Predict house price from many historical examples
- C. Discover customer groups from behaviour
- D. Forecast sales from trends

Training and evaluating on the same rows can mislead because:

- A. Labels are always removed
- **B. It may measure fit to seen data rather than generalisation**
- C. It always converts classification to regression
- D. It automatically creates more data

Which statement about scaling is most accurate?

- A. Every model always requires it
- B. No model can use it
- **C. It helps many models but is not required for every model**
- D. It replaces test evaluation

Part B — Practical Preprocessing Task (40 marks)

Use the customer churn dataset from the Week 4 Day 2 assignment or the dataset supplied by your trainer. Create Week4_Assessment_Preprocessing_YourName.ipynb.

1. Load the data with pandas. Display the first five rows, shape, data types, missing-value counts, duplicate count, and target distribution. (5 marks)
2. Identify the target and input features. Explain why the target must not be included in X. (5 marks)
3. Split into train and test sets before fitting any imputer, scaler, or encoder. Use a fixed random_state and explain the purpose. (5 marks)
4. Build a numeric pipeline with median imputation and scaling. (7 marks)
5. Build a categorical pipeline with missing-value imputation and OneHotEncoder(handle_unknown='ignore'). (7 marks)
6. Combine the steps using ColumnTransformer. Fit on training data only and transform both train and test data. (6 marks)
7. Print transformed shapes and explain what changed. Identify one possible leakage risk. (5 marks)

code:

1. Load and Inspect the Data

```
import pandas as pd

df = pd.DataFrame({
   "age": [24, 35, 29, 42, 31, None, 27, 39, 33, 46, 28, 36],
   "monthly_spend": [500, 1200, None, 2200, 900, None,
                     750, 1800, 1100, 2500, 650, 1500],
   "city": ["Hyderabad", "Chennai", "Hyderabad", "Bengaluru",
            "Chennai", "Hyderabad", "Visakhapatnam", "Bengaluru",
            "Chennai", "Hyderabad", "Bengaluru", "Chennai"],
   "plan_type": ["Basic", "Premium", "Standard", "Premium",
                 "Basic", "Basic", "Standard", "Premium",
                 "Standard", "Premium", "Basic", "Standard"],
   "support_tickets": [2, 0, 1, 3, None, 4, 1, 2, 0, 5, 1, 2],
   "churn": ["Yes", "No", "No", "Yes", "No", "Yes",
             "No", "No", "No", "Yes", "No", "Yes"]
})

print("First five rows:")
print(df.head())
print("\nShape:")
print(df.shape)
print("\nData types:")
print(df.dtypes)
print("\nMissing values:")
print(df.isnull().sum())
print("\nDuplicate rows:")
print(df.duplicated().sum())
print("\nChurn count:")
print(df["churn"].value_counts())
```

Explanation: The dataset has 12 rows and 6 columns. Some numeric values are missing, so we need to fill them.

2. Separate Features and Target

```
X = df[[
   "age", "monthly_spend", "city",
```

```
    "plan_type", "support_tickets"
]]
y = df["churn"]
print("Features shape:", X.shape)
print("Target shape:", y.shape)
```

Explanation: X contains the input data. y contains the churn result we want to predict. We keep the target out of X to avoid giving the model the answer.

3. Split the Data

```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split( X, y, test_size=0.25, random_state=42)
print("Training data:", X_train.shape)
print("Test data:", X_test.shape)
```

Explanation: We use 75% of the data for training and 25% for testing. random_state=42 helps us get the same split each time.

4. Create the Numeric Pipeline

```
from sklearn.pipeline import Pipeline
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import StandardScaler
numeric_features = [
    "age", "monthly_spend", "support_tickets"
]
numeric_pipeline = Pipeline(steps=[
    ("imputer", SimpleImputer(strategy="median")),
    ("scaler", StandardScaler())
])
```

Explanation: We fill missing numbers with the median. Then, we scale the numbers to a similar range.

5. Create the Categorical Pipeline

```
from sklearn.preprocessing import OneHotEncoder
categorical_features = ["city", "plan_type"]
categorical_pipeline = Pipeline(steps=[
    ("imputer", SimpleImputer(strategy="most_frequent")),
    ("encoder", OneHotEncoder(handle_unknown="ignore"))
])
```

Explanation: We fill missing categories with the most common value. Then, we convert text categories into numbers. Unknown categories will not cause an error.

6. Combine the Pipelines

```
from sklearn.compose import ColumnTransformer

preprocessor = ColumnTransformer(transformers=[

  ("num", numeric_pipeline, numeric_features),

  ("cat", categorical_pipeline, categorical_features)

])

X_train_transformed = preprocessor.fit_transform(X_train)

X_test_transformed = preprocessor.transform(X_test)

print("Original training shape:", X_train.shape)

print("Original test shape:", X_test.shape)

print("Transformed training shape:", X_train_transformed.shape)

print("Transformed test shape:", X_test_transformed.shape)
```

Explanation: ColumnTransformer applies the correct steps to each type of column. We fit it on training data and use it to transform both datasets.

7. Explain the Results and Data Leakage

What changed?

- Missing numbers were filled.
- Numeric values were scaled.
- Text categories were converted into separate numeric columns.
- The number of columns may increase after encoding.

Possible data leakage: If we fit the preprocessor on all the data before splitting, test data can influence the results. To avoid this, we split first and fit only on training data.

Part C — Debugging (15 marks)

1. Scaling before splitting: A student scales the full dataset before splitting into train and test.

Your task: Explain the risk and give the correct sequence. (3 marks)

Scaling before splitting can cause data leakage. First, split the data, fit the scaler on training data, and then transform both training and test data.

2. Target leakage: A student includes churn in X and also uses it as y.

Your task: Explain why this invalidates the exercise and correct it. (3 marks)

Including churn in both X and y gives the model the answer. Keep churn only in y.

3. Unknown category: A test row contains a city not seen during training and the encoder raises an error.

Your task: Name the encoder setting that can help and explain it. (3 marks)

Use OneHotEncoder(handle_unknown='ignore'). It handles new categories without raising an error.

4. Wrong imputation: A student fills a numeric column with the text 'unknown' and then scales it.

Your task: Explain the type mismatch and suggest a better strategy. (3 marks)

Using 'unknown' in a numeric column causes an error during scaling. Use SimpleImputer(strategy='median') to fill missing numbers before scaling.

5. Misleading evaluation: A student uses training accuracy as proof the model works on new customers.

Your task: Explain why a held-out test set is needed. (3 marks)

Training accuracy checks data the model has already seen. Use a separate test set to check how well it works on new data.

Part D — Explain in Your Own Words (10 marks)

1. Explain supervised vs. unsupervised learning, with one example. (2 marks)
   Ans: Supervised learning uses labelled data to make predictions. Example: predicting house prices. Unsupervised learning finds patterns in data without labels. Example: grouping customers by shopping habits.

2. Explain features vs. labels using one example. (2 marks)
   Ans: Features are the inputs used to make predictions. The label is what we want to predict. Example: house size and location are features, and house price is the label.

3. Explain why preprocessing should be fitted on training data only. (2 marks)
   Ans: It prevents data leakage and helps us test the model fairly on unseen data.

4. Explain fit vs. transform. (2 marks)
   Ans; fit() learns from the data. transform() uses what it learned to change the data. Example: a scaler learns from training data and then scales the values.

5. Give one problem where a simple rule is preferable to ML. (2 marks)

Ans: ATM PIN verification. The system checks whether the entered PIN is correct, so a simple rule is enough.

Part E — Concept Viva (15 marks)

The trainer will ask you to explain your own notebook. Do not simply read your notes.

- Explain your workflow from loading data to transforming it.

Ans: I load the data using Pandas, check it, split it into training and testing data, and then handle missing values, scale numbers, and encode categories.

- Point to one numeric feature and explain its imputation and scaling.

Ans: I fill missing numeric values using the median. Then, StandardScaler scales the values.

- Print to one categorical feature and explain how encoding changes it.

Ans: OneHotEncoder converts text categories into numeric columns that a machine learning model can understand.

- Explain why the target is excluded from X.

Ans: X contains input features, and y contains the target. We keep the target out of X to prevent data leakage.

- Explain how your code avoids fitting preprocessing on the test set.

Ans: I split the data first and fit preprocessing only on training data. Then, I use it to transform both training and testing data.

- Describe one error you encountered or could encounter and how you would debug it.

Ans: A KeyError happens when a column name is wrong or missing. I check column names using df.columns and correct the name.

Student Submission Checklist

- Quiz answers completed.
- Notebook runs top to bottom.
- Inspection outputs are visible.
- Train/test split happens before fitting transformations.
- Numeric and categorical pipelines are implemented.
- ColumnTransformer is used correctly.
- Debugging and own-words answers are included.
- I can explain my notebook without reading every line.